Xinrui Dong

M.Sc. Student in Probability and Mathematical Statistics

sinzhui.dun@math.msu.ru | +86 186 5388 0441 | Moscow, Russia

Research Interests

- Branching processes and catalytic branching random walks; statistical inference for branching populations
- Stochastic and Markov processes; limit theorems; ergodic methods in mathematical finance

Education

Lomonosov Moscow State University (MSU) Sep 2025 - Jul 2027 (expected)

M.Sc. in Mathematics, Probability and Mathematical Statistics Moscow, Russia

- Faculty of Mechanics and Mathematics; Department of Mathematical Statistics and Stochastic Processes
- Academic advisor: Associate Professor Ekaterina V. Bulinskaya
- Current research: catalytic branching random walks; a constructive propagation-front project as preparation for the Master's thesis
- Theoretical preparation: real analysis, measure theory, martingales, renewal theory, large deviations and multitype branching processes

Lomonosov Moscow State University, Institute of Russian Language and Culture Sep 2024 - Jun 2025

Preparatory Programme in Russian, Economics Track Moscow, Russia

- Completed an intensive one-year Russian programme and received an MSU Russian Language Proficiency Certificate
- Supplementary mathematical study included Mathematical Analysis and Financial Mathematics

Xiamen University Tan Kah Kee College Sep 2020 - Jul 2024

Bachelor of Engineering (B.Eng.) Zhangzhou, China

- Selected quantitative coursework: Calculus (Higher Mathematics), Linear Algebra, Probability and Statistics, Mathematical Modelling

Selected Research Projects

Course Paper: From the Galton-Watson Process to Catalytic Branching Random Walks 2026

Expository review; grade: 5/5; supervised by Associate Professor Ekaterina V. Bulinskaya

- Developed a structured review linking the Galton-Watson process to spatially inhomogeneous branching models on the integer lattice
- Reviewed catalyst-dependent dynamics, Perron-root criticality and propagation-front results under Cramér's condition
- Current follow-on project: constructing explicit catalytic branching models with prescribed propagation-front geometries, following worked examples in my supervisor's research

Long-Term Asset Allocation through Ergodic Theory Oct 2025 - Feb 2026

- Independently developed a continuous-time stochastic market and self-financing portfolio model using time-homogeneous Markov dynamics
- Applied Foster-Lyapunov and ergodic arguments to study invariant measures, time-average convergence and long-run portfolio growth
- Showed how stationary market characteristics determine asymptotic allocation and average capital growth under the model assumptions; the resulting article was published in 2026

Publication

- **Dong, X.** (2026). Study of a Long-Term Asset Allocation Model from an Ergodic Theory Perspective. *Innovation & Investment*, no. 3. Sole-authored; published in a peer-reviewed journal included in the Russian Higher Attestation Commission (VAK) list. [In Russian]

Honours and Awards

First Prize, 2nd National University Students' Mathematics Ability Challenge Jan 2025

MSU Russian Language Proficiency Certificate Jun 2025

Technical and Language Skills

- **Methods:** branching and Markov processes, stochastic differential equations, ergodic theory and proof reconstruction
- **Programming and tools:** Python (undergraduate coursework; refreshing stochastic simulation skills), LaTeX and Excel
- **Languages:** Chinese (native); Russian (MSU certificate); English (academic reading; preparing for IELTS Academic)